

es. (visto nell' lezione precedente)

$$U \sim U(0, 1)$$

$$Y = -\ln U$$

si verifica che  $Y \sim E(1)$

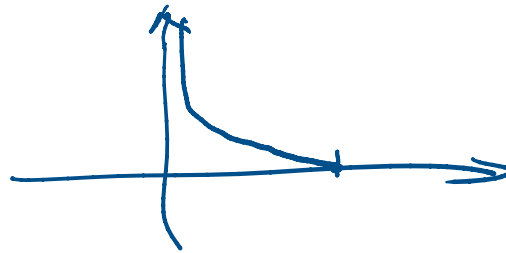
$$f_Y(a) = f_X(\bar{g}'(a)) \left| \frac{d\bar{g}'(a)}{da} \right|$$

$$a = g(u) = -\ln u \Rightarrow u = e^{-a} = \bar{g}'(a)$$

$$\left| \frac{d\bar{g}'(a)}{da} \right| = |-e^{-a}| = e^{-a}$$

$$f_Y(a) = \begin{cases} 1 & \text{se } \bar{g}'(a) \in [0, 1] \\ 0 & \text{altrove} \end{cases} \cdot e^{-a} = \begin{cases} e^{-a} & \text{se } e^{-a} \in [0, 1] \\ 0 & \text{altrove} \end{cases} =$$

$$= \begin{cases} e^{-a} & \text{se } a \geq 0 \\ 0 & \text{altrove} \end{cases}$$



Come modificare l'esercizio per ottenere  
 $T \sim E(\lambda) \quad \lambda \in \mathbb{R}^+$

$$cY \sim E\left(\frac{1}{c}\right) \quad \frac{1}{c} = \lambda \Rightarrow c = \frac{1}{\lambda}$$

$$\Downarrow$$
$$T = -\frac{1}{\lambda} \ln U \sim E(\lambda)$$

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## FUNZIONE DI DUE V.C. CONGIUNTAMENTE CONTINUE NOTE

$(X, Y)$  v.c. continue note ( $f(x, y)$  nota)

$$Z = h(X, Y)$$

$$E[Z] = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h(x, y) f(x, y) dx dy$$

$$E[Z^2] = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h^2(x, y) f(x, y) dx dy$$

$$E[Z^2] = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h^2(x, y) f(x, y) dx dy$$

$$\text{Var}(Z) = E[Z^2] - E[Z]^2$$

$$F_Z(a) = P(Z \leq a) = P(h(X, Y) \leq a) = \dots$$

$$f_Z(a) = \frac{dF_Z(a)}{da}$$

**CASO PARTICOLARE**  $h(X, Y) = X + Y$

$$\begin{aligned} F_Z(a) &= P(X + Y \leq a) = P(Y \leq -X + a) = \\ &= \int_{-\infty}^{+\infty} \left( \int_{-\infty}^{-x+a} f(x, y) dy \right) dx \end{aligned}$$

$$\underline{f_Z(a)} = \frac{dF_Z(a)}{da} = \frac{d}{da} \left( \int_{-\infty}^{+\infty} \left( \int_{-\infty}^{-x+a} f(x, y) dy \right) dx \right) =$$

$$= \int_{-\infty}^{+\infty} f(x, -x+a) dx$$

oppure

$$f_z(a) = \int_{-\infty}^{+\infty} f(-y+a, y) dy$$

Se  $X$  e  $Y$  sono V.C. INDIPENDENTI

$$f_z(a) = \int_{-\infty}^{+\infty} f_x(x) f_y(-x+a) dx$$

oppure equivalentemente

$$f_z(a) = \int_{-\infty}^{+\infty} f_x(-y+a) f_y(y) dy$$

SOMMA DI DUE V.C. UNIFORMI INDIPENDENTI

es. 1  $X_1 \sim U(0,1)$ ,  $X_2 \sim U(0,1)$  INDIP.

✓ ✓ ✓

1+1=2

es. 1.  $X_1 \sim U(0,1)$ ,  $X_2 \sim U(0,1)$  ...

$$Y = X_1 + X_2$$

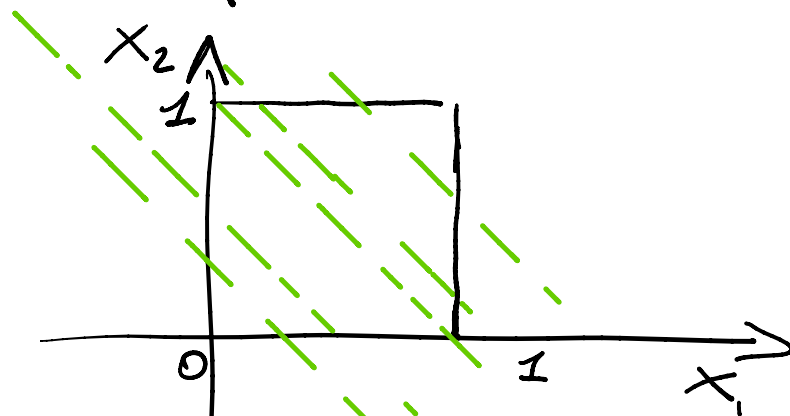
$$E[X_k] = \frac{1}{2}, \text{Var}(X_k) = \frac{1}{12}$$

$$E[Y] = E[X_1 + X_2] = E[X_1] + E[X_2] = \frac{1}{2} + \frac{1}{2} = 1$$

$$\text{Var}(Y) = \text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2) = \frac{1}{12} + \frac{1}{12} = \frac{1}{6}$$

$$F_Y(a) = P(X_1 + X_2 \leq a) = \begin{cases} 0 & \text{se } a < 0 \\ 1 & \text{se } a > 2 \end{cases}$$

$$F_Y(a) = P(X_2 \leq -X_1 + a)$$

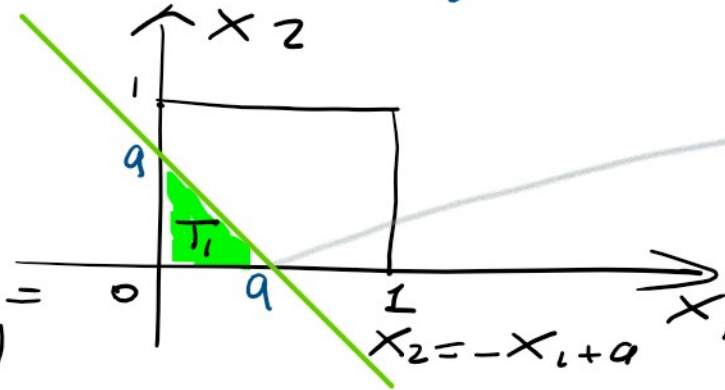


Si distingue il caso  $a \in [0,1]$  dal caso  $a \in ]1,2]$   
 dato che per  $a=1$  si ha l'ultimo caso in cui  
 l'intersezione tra il semipiano  $X_2 \leq -X_1 + a$   
 e il quadrato di def. è un triangolo

l'intersezione  $1/2$  "semiprimo"  $\wedge 2 \leq -x_1 + a$   
 e il quadrato di def.  $\bar{c}$  un triangolo

Se  $a \in [0, 1]$

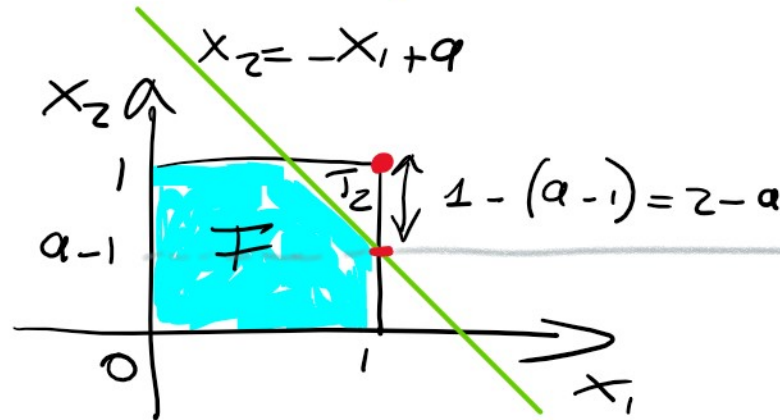
$$\begin{aligned} P(X_2 \leq -X_1 + a) &= \\ &= \frac{\text{Area}(\mathcal{T}_1)}{\text{Area}(\text{Quadrato di def.})} = \\ &= \frac{\frac{a \cdot a}{2}}{1^2} = \frac{a^2}{2} \end{aligned}$$



$$\begin{cases} x_2 = -x_1 + a \\ x_2 = 0 \end{cases}$$

Se  $a \in ]1, 2]$

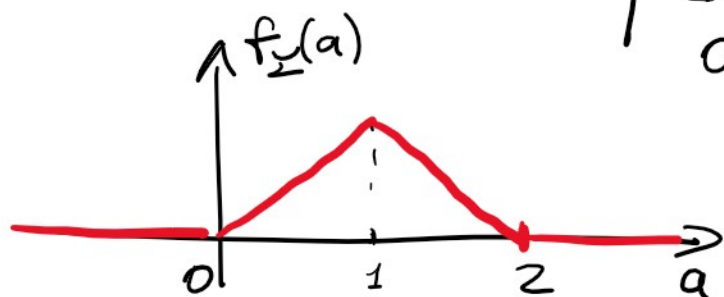
$$\begin{aligned} P(X_2 \leq -X_1 + a) &= \\ &= \frac{\text{Area}(\mathcal{F})}{\text{Area}(\text{Quadrato di def.})} = \frac{\text{Area}(\text{Quadrato di def.}) - \text{Area}(\mathcal{T}_2)}{\text{Area}(\text{Quadrato di def.})} = \\ &= \frac{1^2 - \frac{(2-a)^2}{2}}{1^2} = 1 - \frac{(2-a)^2}{2} \end{aligned}$$



$$\begin{cases} x_2 = -x_1 + a \\ x_1 = 1 \end{cases} \Rightarrow x_2 = a - 1$$

Risumendo  $F_Y(a) = \begin{cases} 0 & \text{se } a < 0 \\ \frac{a^2}{2} & \text{se } 0 \leq a \leq 1 \\ 1 - \frac{(2-a)^2}{2} & \text{se } 1 < a \leq 2 \\ 1 & \text{se } a > 2 \end{cases}$

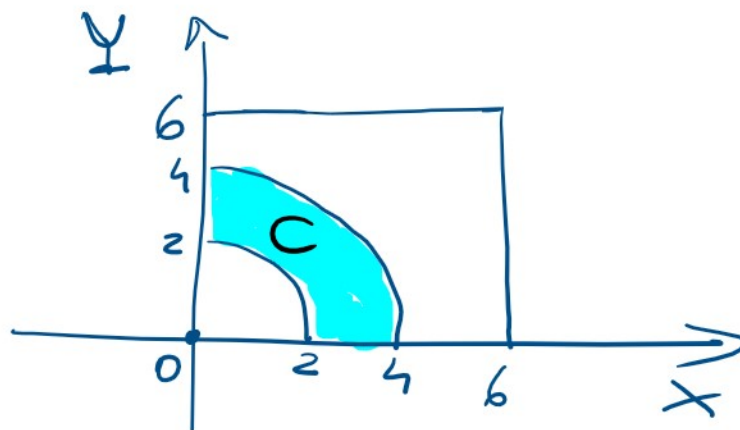
$$f_Y(a) = \frac{dF_Y(a)}{da} = \begin{cases} 0 & \text{se } a < 0 \\ a & \text{se } 0 \leq a \leq 1 \\ 2-a & \text{se } 1 < a \leq 2 \\ 0 & \text{se } a > 2 \end{cases}$$

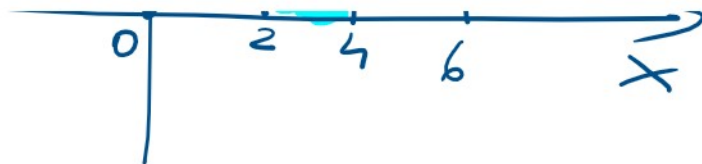


Q4 del 18/12/2013

$$\left. \begin{aligned} X &\sim U(0,6) \\ Y &\sim U(0,6) \end{aligned} \right\} \text{INDIP.}$$

P... ..





$$f(x, y) = f_X(x) f_Y(y) =$$

$$= \begin{cases} \frac{1}{6} & \text{se } x \in [0, 6] \\ 0 & \text{altrove} \end{cases} \begin{cases} \frac{1}{6} & \text{se } y \in [0, 6] \\ 0 & \text{altrove} \end{cases} = \begin{cases} \frac{1}{36} & \text{per } x, y \in [0, 6] \\ 0 & \text{altrove} \end{cases}$$

$$P((X, Y) \in C) = \frac{\text{Area}(C)}{\text{Area}(\text{Quadrato di def.})} = \frac{\frac{1}{4}(\pi 4^2 - \pi 2^2)}{36} =$$

$$= \frac{\frac{\pi}{4}(16-4)}{36} = \frac{\pi}{12}$$

$$Z = 2X + 2Y$$

$$X_1 = 2X \sim U(0, 12)$$

$$Y_1 = 2Y \sim U(0, 12)$$

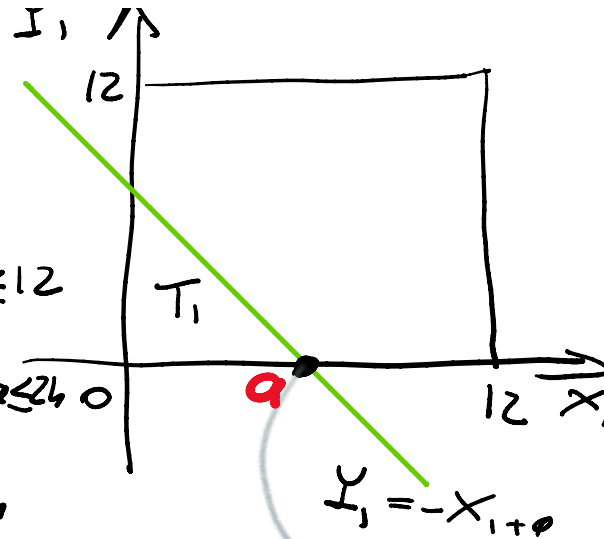
... ..

per proprietà dell  
trasformazione lineare  
di una v.c. Uniforme  
viste ieri.



$$Z = X_1 + Y_1$$

$$F_Z(a) = \begin{cases} 0 & \text{se } a < 0 \\ & \text{se } 0 \leq a \leq 12 \\ 1 & \text{se } a > 24 \end{cases}$$



caso  $a \in [0, 12]$

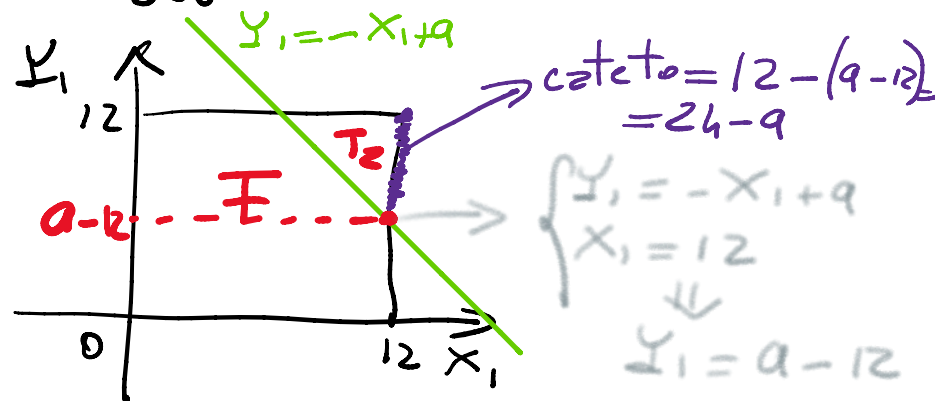
$$F_Z(a) = P(Y_1 \leq -X_1 + a) = \frac{\text{Area } T_1}{\text{Area}(\text{Quadrato di def.})} =$$

$$= \frac{\frac{a \cdot a}{2}}{144} = \frac{a^2}{288}$$

caso  $a \in [12, 24]$

$$F_Z(a) = P(Y_1 \leq -X_1 + a) =$$

$$= \frac{\text{Area}(F)}{\text{Area}(\text{Quadrato di def.})} =$$



$$= \frac{\text{Aree}(\text{Quadrato di def.}) - \text{Aree}(\Gamma_2)}{\text{Aree}(\text{Quadrato di def.})} =$$

$$= \frac{144 - \frac{1}{2}(24-a)^2}{144} = 1 - \frac{(24-a)^2}{288}$$

$$F_Z(a) = \begin{cases} 0 & \text{se } a < 0 \\ \frac{a^2}{288} & \text{se } 0 \leq a \leq 12 \\ 1 - \frac{(24-a)^2}{288} & \text{se } 12 < a \leq 24 \\ 1 & \text{se } a > 24 \end{cases}$$

esempio  $X \sim U(0,2); Y \sim U(0,1)$  INDIP.

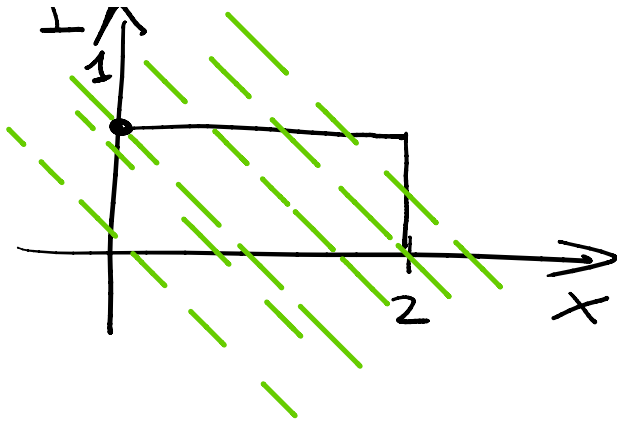
$$Z = X + Y$$

$$E[Z] = E[X] + E[Y] = 1 + \frac{1}{2} = \frac{3}{2}$$

$$\text{Var}(Z) = \underset{\substack{\uparrow \\ \text{INDIP.}}}{\text{Var}(X)} + \text{Var}(Y) = \frac{2^2}{12} + \frac{1^2}{12} = \frac{5}{12}$$

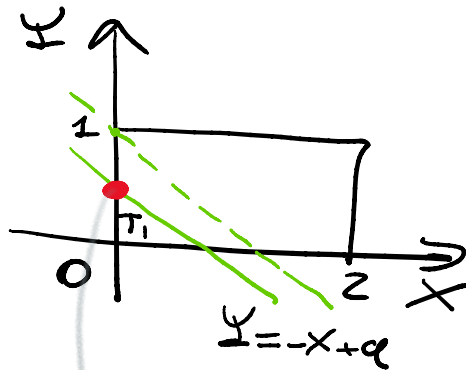


F(1) P(X, Y) = ...



$$F_2(a) = P(X+Y \leq a) = P(Y \leq -X+a) =$$

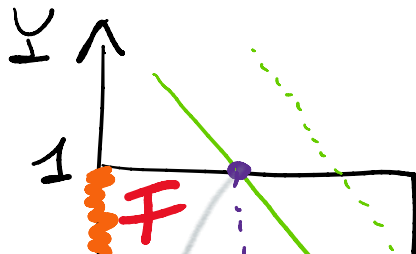
$$= \begin{cases} 0 & \text{se } a < 0 \\ 1 & \text{se } a > 3 \end{cases}$$



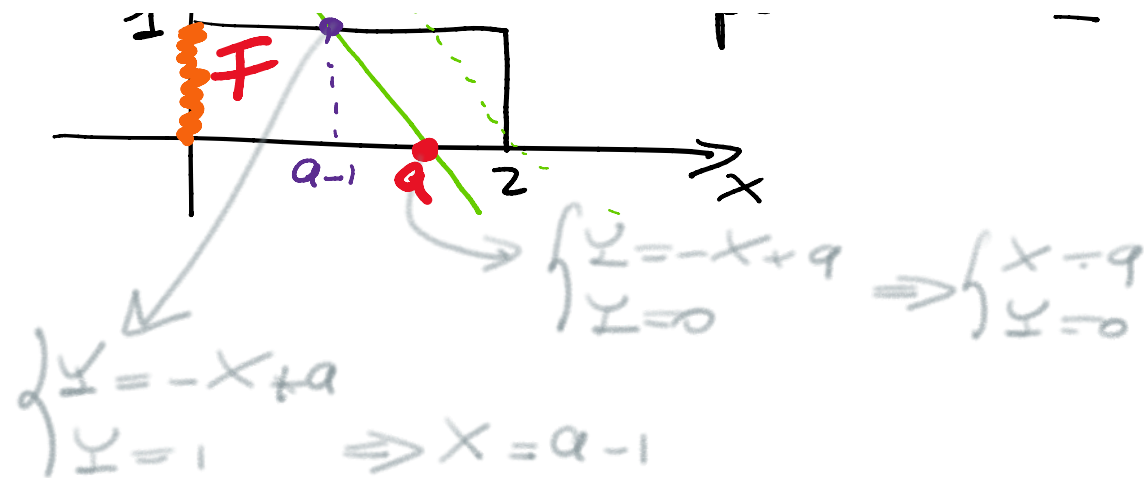
per  $a \in [0, 1]$

$$\begin{cases} Y = -X+a \\ X=0 \end{cases} \Rightarrow \begin{matrix} Y=a \\ X=0 \end{matrix}$$

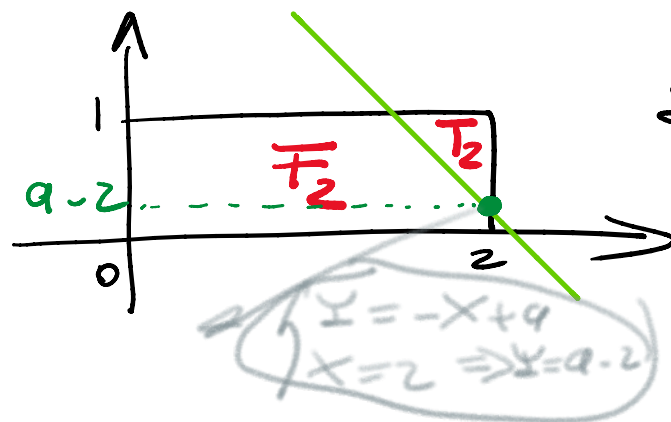
$$\begin{aligned} F_2(a) &= P(Y \leq -X+a) = \\ &= \frac{\text{Area}(\tau_1)}{\text{Area}(\text{Rettangolo di def.})} = \\ &= \frac{\frac{a \cdot a}{2}}{1 \cdot 2} = \frac{a^2}{4} \end{aligned}$$



per  $1 < a \leq 2$



$$\begin{aligned}
 F_2(a) &= P(Y \leq -X + a) = \frac{\text{Area}(F)}{\text{Area}(\text{Rettangolo di def.})} = \\
 &= \frac{\frac{(a + (a-1)) \cdot 1}{2}}{1 \cdot 2} = \frac{2a-1}{4}
 \end{aligned}$$



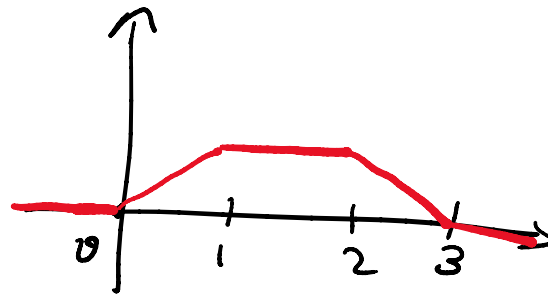
Se  $a \in ]2, 3]$

$$\begin{aligned}
 F_2(a) &= P(Y \leq -X + a) = \\
 &= \frac{\text{Area}(F_2)}{\text{Area}(\text{Rettangolo di def.})} = \\
 &= \frac{\text{Area}(\text{Rettangolo di def.}) - \text{Area}(T_2)}{\text{Area}(\text{Rettangolo di def.})} =
 \end{aligned}$$

$$\overline{T}_2(a) = \frac{1 \cdot 2 - \frac{(3-a)^2}{2}}{1 \cdot 2} = 1 - \frac{(3-a)^2}{4}$$

$$\overline{T}_2(a) = \begin{cases} 0 & \text{se } a < 0 \\ \frac{a^2}{4} & \text{se } 0 \leq a \leq 1 \\ \frac{2a-1}{4} & \text{se } 1 < a \leq 2 \\ 1 - \frac{(3-a)^2}{4} & \text{se } 2 < a \leq 3 \\ 1 & \text{se } a > 3 \end{cases}$$

$$f_2(a) = \begin{cases} 0 & \text{se } a < 0 \\ \frac{a}{2} & \text{se } 0 \leq a \leq 1 \\ \frac{1}{2} & \text{se } 1 < a \leq 2 \\ \frac{3-a}{2} & \text{se } 2 < a \leq 3 \\ 0 & \text{se } a > 3 \end{cases}$$



TEOREMA DEL LIMITE STABILE

## TEOREMA DEL LIMITE CENTRALE O CENTRALE DEL LIMITE

Dato una successione di v.c. i.i.d.  $X_1, X_2, \dots, X_n$   
di valor medio  $E[X_k] = \mu$  per  $k=1, \dots, n$  e  
di varianza  $Var(X_k) = \sigma^2$  per  $k=1, \dots, n$ ,

allora  $Y_n = \frac{\sum_{k=1}^n X_k - n\mu}{\sigma\sqrt{n}}$  converge in

distribuzione ad una normale standard per  $n \rightarrow +\infty$ ,  
ovvero

$$\forall a \in \mathbb{R} \quad \lim_{n \rightarrow +\infty} P(Y_n \leq a) = \Phi_2(a) \quad \text{con } Z \sim N(0,1)$$

### OSS 1

$$Y_n = \frac{\sum_{k=1}^n X_k - n\mu}{\sigma\sqrt{n}}$$

$$E[Y_n] = E\left[\frac{\sum_{k=1}^n X_k - n\mu}{\sigma\sqrt{n}}\right] = \frac{1}{\sigma\sqrt{n}} E\left[\sum_{k=1}^n X_k - n\mu\right] =$$

$$\begin{aligned}
 E[\bar{X}_n] &= E\left[\frac{\sum_{k=1}^n X_k - n\mu}{\sigma\sqrt{n}}\right] = \frac{1}{\sigma\sqrt{n}} E\left[\sum_{k=1}^n X_k - n\mu\right] = \\
 &= \frac{1}{\sigma\sqrt{n}} \left\{ \sum_{k=1}^n E[X_k] - E[n\mu] \right\} = \frac{1}{\sigma\sqrt{n}} \left\{ \left(\sum_{k=1}^n \mu\right) - n\mu \right\} = \\
 &= \frac{1}{\sigma\sqrt{n}} \{n\mu - n\mu\} = 0
 \end{aligned}$$

$$\begin{aligned}
 Var(\bar{X}_n) &= Var\left(\frac{\sum_{k=1}^n X_k - n\mu}{\sigma\sqrt{n}}\right) = \\
 &= Var\left(\frac{1}{\sigma\sqrt{n}} \sum_{k=1}^n X_k - \frac{n\mu}{\sigma\sqrt{n}}\right) = \left(\frac{1}{\sigma\sqrt{n}}\right)^2 Var\left(\sum_{k=1}^n X_k\right) = \\
 &= \frac{1}{n\sigma^2} \left(\sum_{k=1}^n Var(X_k)\right) = \frac{1}{n\sigma^2} \sum_{k=1}^n \sigma^2 = \frac{n\sigma^2}{n\sigma^2} = 1
 \end{aligned}$$

**OSS. Ambiti di applicazione del TLC**

$$1) T \sim B(n, p) \quad T = \sum_{k=1}^n X_k \quad \text{con } X_k \underset{\text{INDIP.}}{\sim} Be(p)$$

per  $T$  al crescere di  $n$  è ragionevole

per  $T$  al crescere di  $n$  è ragionevole considerare l'applicazione del TLC

$$\text{per } n \gg 1 \quad Y_n = \frac{T - n\mu}{\sigma\sqrt{n}} \sim N(0,1)$$

$$\Rightarrow T \sim N(n\mu, \sigma\sqrt{n})$$

$$E[X_k] = p, \quad \text{Var}(X_k) = pq \quad (\text{Bernoulli})$$

$$\Rightarrow T \sim N(np, \sqrt{npq})$$

Negli esercizi questa appross. si può considerare a partire da  $n \geq 30$

$$\text{es. } n=100 \quad T \sim B(100, p)$$

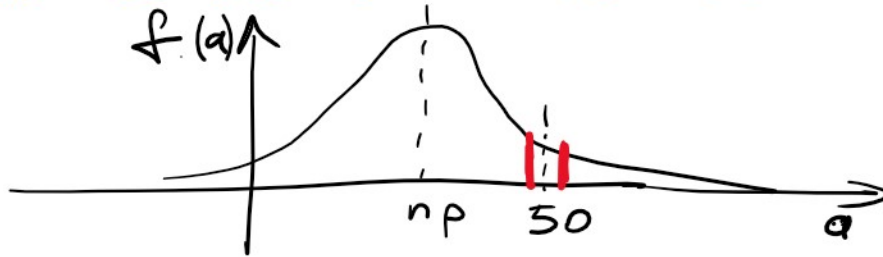
$$P(T=50) = \binom{100}{50} p^{50} (1-p)^{100-50}$$

$$\text{se invece } T \sim N(100p, \sqrt{100p(1-p)})$$

$$P(T=50)$$

$$P(T=50) = 0$$

METODO DI APPROSS ALLA CONTINUITA'



$$\begin{aligned}
 P(T=50) &= P\left(50 - \frac{1}{2} < T \leq 50 + \frac{1}{2}\right) = \\
 &= F_T\left(50 + \frac{1}{2}\right) - F_T\left(50 - \frac{1}{2}\right) \approx z \sim N(0,1) \\
 &= P\left(\frac{50 - \frac{1}{2} - np}{\sqrt{np(1-p)}} < \frac{T - np}{\sqrt{np(1-p)}} \leq \frac{50 + \frac{1}{2} - np}{\sqrt{np(1-p)}}\right) = \\
 &= P\left(\frac{49.5 - np}{\sqrt{np(1-p)}} < z < \frac{50.5 - np}{\sqrt{np(1-p)}}\right) = \\
 &= F_z\left(\frac{50.5 - np}{\sqrt{np(1-p)}}\right) - F_z\left(\frac{49.5 - np}{\sqrt{np(1-p)}}\right)
 \end{aligned}$$

$$\left( \frac{1}{\sqrt{np(1-p)}} \right) \approx \left( \frac{1}{\sqrt{np(1-p)}} \right)$$

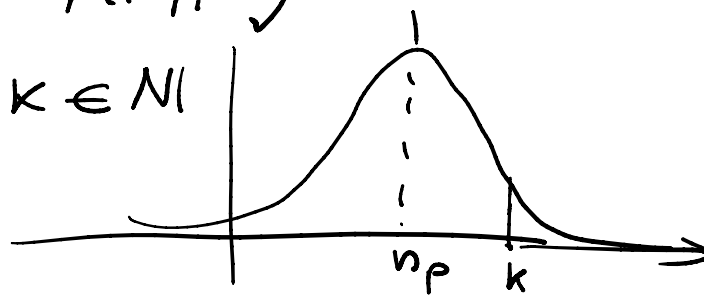
IN GENERALE DATO  $k \in \mathbb{N}$

$$P(T > k) = P(T \geq k + \frac{1}{2})$$

$$P(T \geq k) = P(T \geq k - \frac{1}{2})$$

$$P(T < k) = P(T < k - \frac{1}{2})$$

$$P(T \leq k) = P(T \leq k + \frac{1}{2})$$



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$$X_k \text{ con } f(x) = \begin{cases} cx^2 & \text{se } x \in [0, 1] \\ 0 & \text{altrove} \end{cases}$$

$$f(x) \geq 0 \Rightarrow c \geq 0$$

$$1 = \int_{-\infty}^{+\infty} f(x) dx = \int_0^1 cx^2 dx = c \left[ \frac{x^3}{3} \right]_0^1 = \frac{c}{3}$$

$$\Rightarrow c = 3$$

$$P(X_k \leq 1) = \int_0^1 3x^2 dx = \left[ x^3 \right]_0^1 = 1$$

$$P(X_k \leq \frac{1}{2}) = \int_0^{\frac{1}{2}} 3x^2 dx = \left[ \frac{3x^3}{3} \right]_0^{\frac{1}{2}} = \frac{1}{8}$$

$Y$  = 'no di dispositivi da sostituire entro 6 mesi su 100 usati'

$$Y \sim B(100, \frac{1}{8})$$

$$P(Y \geq 50) = \sum_{k=50}^{100} \binom{100}{k} \left(\frac{1}{8}\right)^k \left(\frac{7}{8}\right)^{100-k}$$

per TLC  $Y \sim N\left(\frac{25}{2}, \sqrt{\frac{25 \cdot 7}{2 \cdot 8}}\right) = N\left(\frac{25}{2}, \frac{5}{4}\sqrt{7}\right)$

$$P(Y \geq 50) = P(Y \geq 49.5) = P\left(\frac{Y - \frac{25}{2}}{\frac{5}{4}\sqrt{7}} \geq \frac{49.5 - \frac{25}{2}}{\frac{5}{4}\sqrt{7}}\right) =$$

$$\approx P\left(Z \geq \frac{37}{\frac{5}{4}\sqrt{7}}\right) = 1 - F_Z\left(\frac{37}{\frac{5}{4}\sqrt{7}}\right)$$

2)  $\bar{X} = \sum_{k=1}^n X_k$  media aritmetica delle  $X_k$

2)  $\bar{X} = \sum_{k=1}^n \frac{X_k}{n}$  media aritmetica delle  $X_k$   
 (detta media campionaria  
 in statistica)

$X_1, \dots, X_n$  i.i.d.

$$Y_n = \frac{\sum_{k=1}^n X_k - n\mu}{\sigma\sqrt{n}} \stackrel{\substack{\text{divisione} \\ \text{per } n}}{=} \frac{\sum_{k=1}^n \frac{X_k}{n} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$\lim_{n \rightarrow \infty} \Phi\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \leq a\right) = F_Z(a) \quad \forall a \in \mathbb{R}$$

$Z \sim N(0,1)$

Ovvero se  $n \gg 1$

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

es. Q5 18/12/2013

$$X_k \quad E[X_k] = 20, \quad \sqrt{\text{Var}(X_k)} = 3$$

$$\bar{X}_{36} \sim N\left(20, \frac{3}{\sqrt{36}}\right) = N\left(20, \frac{1}{2}\right)$$

$$\bar{X}_{49} \sim N\left(20, \frac{3}{\sqrt{49}}\right) = N\left(20, \frac{3}{7}\right)$$

$$1) P(\bar{X}_{49} > \bar{X}_{36})? \quad 2) P(\bar{X}_{36} \leq 25)?$$

$$3) P(|\bar{X}_{49} - \bar{X}_{36}| < 3)$$

$$\begin{aligned} 1) P(\bar{X}_{49} > \bar{X}_{36}) &= P(\bar{X}_{49} - \bar{X}_{36} > 0) = \\ &= P(D > 0) = P\left(\frac{D - 0}{\sqrt{\frac{9}{49} + \frac{1}{4}}} > \frac{0 - 0}{\sqrt{\frac{9}{49} + \frac{1}{4}}}\right) = \\ &= P(Z > 0) = \frac{1}{2} \end{aligned}$$

$$2) P(\bar{X}_{36} \leq 25) = P\left(\frac{\bar{X}_{36} - 20}{\frac{1}{2}} \leq \frac{25 - 20}{\frac{1}{2}}\right) = P(Z \leq 10) \approx 1$$

$$\left\{ \begin{array}{l} \bar{X}_{49} \sim N\left(20, \frac{3}{7}\right) \\ - \bar{X}_{36} \sim N\left(-20, \frac{1}{2}\right) \\ D = \bar{X}_{49} - \bar{X}_{36} \sim \\ \sim N\left(0, \sqrt{\frac{9}{49} + \frac{1}{4}}\right) \end{array} \right.$$

$$\begin{aligned}
3) \quad P(|\bar{X}_{49} - \bar{X}_{36}| < 3) &= P(-3 < \bar{X}_{49} - \bar{X}_{36} < 3) = \\
&= P(-3 < D < 3) = P\left(\frac{-3-0}{\sqrt{\frac{9}{49} + \frac{1}{4}}} < \frac{D-0}{\sqrt{\frac{9}{49} + \frac{1}{4}}} < \frac{3-0}{\sqrt{\frac{9}{49} + \frac{1}{4}}}\right) = \\
&= P\left(-\frac{3}{\sqrt{\frac{9}{49} + \frac{1}{4}}} < Z < \frac{3}{\sqrt{\frac{9}{49} + \frac{1}{4}}}\right) = F_2\left(\frac{3}{\sqrt{\frac{9}{49} + \frac{1}{4}}}\right) - F_2\left(\frac{-3}{\sqrt{\frac{9}{49} + \frac{1}{4}}}\right) = \\
&= F_2\left(\frac{3}{\sqrt{\frac{9}{49} + \frac{1}{4}}}\right) - \left(1 - F_2\left(\frac{3}{\sqrt{\frac{9}{49} + \frac{1}{4}}}\right)\right) = \\
&= 2F_2\left(\frac{3}{\sqrt{\frac{9}{49} + \frac{1}{4}}}\right) - 1
\end{aligned}$$

3)  $X_1, X_2, \dots, X_n$  i.i.d.

$$Y_n = \frac{\sum_{k=1}^n X_k - n\mu}{\sigma\sqrt{n}} \sim N(0, 1) \quad \text{as } n \gg 1$$

$$\Rightarrow \sum_{i=1}^n X_i \sim N(n\mu, \sigma\sqrt{n})$$

$$\Rightarrow \sum_{k=1}^n X_k \sim N(n\mu, \sigma\sqrt{n})$$

$$E\left[\sum_{k=1}^n X_k\right] = \sum_{k=1}^n E[X_k] = n\mu$$

$$\text{Var}\left(\sum_{k=1}^n X_k\right) \underset{\substack{\uparrow \\ \text{INDIP.}}}{=} \sum_{k=1}^n \text{Var}(X_k) = n\sigma^2$$

## DIMOSTRAZIONE DEL TLC

Consideriamo il caso particolare di  $\mu=0$  e  $\sigma=1$   
IPOTESI

$X_1, \dots, X_n$  iid  $E[X_k]=0, \text{Var}(X_k)=1$

$$Y_n = \frac{\sum_{k=1}^n X_k - n \cdot 0}{1 \cdot \sqrt{n}} = \frac{\sum_{k=1}^n X_k}{\sqrt{n}}$$

TESI  $\lim_{n \rightarrow +\infty} P(Y_n \leq a) = \Phi_2(a)$

$$\lim_{n \rightarrow +\infty} P(Y_n \leq a) = \Phi_Z(a)$$

$$\text{ovvero } \lim_{n \rightarrow +\infty} \phi_{Y_n}(t) = e^{\frac{t^2}{2}}$$

$$\text{ovvero } \lim_{n \rightarrow +\infty} \ln(\phi_{Y_n}(t)) = \frac{t^2}{2}$$

Definizione

$$L(t) = \ln(\phi_{X_\kappa}(t))$$

$$L(0) = \ln(\phi_{X_\kappa}(0)) = \ln(1) = 0$$

$$L'(t) = \frac{d}{dt} L(t) = \frac{d}{dt} (\ln(\phi_{X_\kappa}(t))) = \frac{1}{\phi_{X_\kappa}(t)} \frac{d\phi_{X_\kappa}(t)}{dt}$$

$$L'(0) = \frac{1}{1} \left. \frac{d\phi_{X_\kappa}(t)}{dt} \right|_{t=0} = E[X_\kappa] = 0$$

$$L''(t) = \frac{d^2 L(t)}{dt^2} = \frac{d}{dt} \left( \frac{1}{\phi_{X_K}(t)} \frac{d\phi_{X_K}(t)}{dt} \right) =$$

$$= - \left( \frac{1}{\phi_{X_K}^2(t)} \frac{d\phi_{X_K}(t)}{dt} \right) \frac{d\phi_{X_K}(t)}{dt} + \frac{1}{\phi_{X_K}(t)} \frac{d^2 \phi_{X_K}(t)}{dt^2}$$

$$L''(0) = - \frac{1}{1^2} \left( \left. \frac{d\phi_{X_K}(t)}{dt} \right|_{t=0} \right)^2 + \frac{1}{1} \left. \frac{d^2 \phi_{X_K}(t)}{dt^2} \right|_{t=0} =$$

$$= - E[X_K^2] + E[X_K^2] = \text{Var}(X_K) = 1$$

$$\phi_{Y_n}(t) = \phi_{\sum_{k=1}^n X_k}(t) \stackrel{\text{INDIP.}}{=} \phi_{\frac{X_1}{\sqrt{n}}}(t) \phi_{\frac{X_2}{\sqrt{n}}}(t) \dots \phi_{\frac{X_n}{\sqrt{n}}}(t) =$$

$$= \phi_{X_1}\left(\frac{t}{\sqrt{n}}\right) \phi_{X_2}\left(\frac{t}{\sqrt{n}}\right) \dots \phi_{X_n}\left(\frac{t}{\sqrt{n}}\right) \stackrel{\text{i.i.d.}}{=} \phi_{X_1}(t) = \phi_{X_n}(t)$$

$$= \left( \phi_{X_1}\left(\frac{t}{\sqrt{n}}\right) \right)^n$$

$$\phi_{\frac{X_1}{\sqrt{n}}}(t) =$$

$$= E[e^{t \frac{X_1}{\sqrt{n}}}] =$$

$$= E[e^{\frac{t}{\sqrt{n}} X_1}] =$$

$$= \phi_{X_1}\left(\frac{t}{\sqrt{n}}\right)$$

$(1 + \frac{t}{\sqrt{n}})$

$= \phi_{x_1}(\frac{t}{\sqrt{n}})$

$$\ln \phi_{\bar{X}_n}(t) = \ln \left[ \left( \phi_{x_1}(\frac{t}{\sqrt{n}}) \right)^n \right] = n \ln \left( \phi_{x_1}(\frac{t}{\sqrt{n}}) \right) = n L\left(\frac{t}{\sqrt{n}}\right)$$

$$\begin{aligned} \lim_{n \rightarrow +\infty} \ln \left( \phi_{\bar{X}_n}(t) \right) &= \lim_{n \rightarrow +\infty} n L\left(\frac{t}{\sqrt{n}}\right) = \\ &= \lim_{n \rightarrow +\infty} \frac{L\left(\frac{t}{\sqrt{n}}\right)}{n^{-1}} = \lim_{n \rightarrow +\infty} \frac{L'\left(\frac{t}{\sqrt{n}}\right) \left(-\frac{1}{2} t n^{-\frac{3}{2}}\right)}{-n^{-2}} = \\ &= \lim_{n \rightarrow +\infty} \frac{\frac{t}{2} L'\left(\frac{t}{\sqrt{n}}\right)}{n^{-\frac{1}{2}}} = \lim_{n \rightarrow +\infty} \frac{\frac{t}{2} L''\left(\frac{t}{\sqrt{n}}\right) \left(-\frac{1}{2} t n^{-\frac{3}{2}}\right)}{-\frac{1}{2} n^{-\frac{3}{2}}} = \\ &= \lim_{n \rightarrow +\infty} \frac{t^2}{2} L''\left(\frac{t}{\sqrt{n}}\right) = \frac{t^2}{2} \end{aligned}$$

Se  $\mu \neq 0$  e  $\sigma \neq 1$        $\bar{X}_n = \frac{\sum_{k=1}^n X_k - n\mu}{\sigma\sqrt{n}}$

$$X'_k = \frac{X_k - \mu}{\sigma} \Rightarrow Y_n = \sum_{k=1}^n \frac{X'_k}{\sqrt{n}}$$

$$E[X'_k] = \frac{E[X_k] - E[\mu]}{\sigma} = 0; \quad \text{Var}(X'_k) = \frac{1}{\sigma^2} \text{Var}(X_k) = 1$$

$X'_k$  i.i.d.

□

Q6 DICEMBRE 2014

13 lampadine 7 nuove

$U_k = \text{lampadina usata}$   
 $N_k = \text{"}$

$H_1$  estrazione di  $U_1, U_2$  il I giorno

$H_2$  " di  $U_1$  e  $N_1$  il I giorno

$H_3$  " di  $N_1$  e  $N_2$  il I giorno

$$P(H_1) = \frac{6 \cdot 5}{13 \cdot 12}$$

$$P(H_3) = \frac{7 \cdot 6}{13 \cdot 12}$$

$$P(H_3) = \frac{7 \cdot 6}{13 \cdot 12}$$

$$P(H_2) = \frac{7 \cdot 6 + 6 \cdot 7}{13 \cdot 12} = \frac{2 \cdot 6 \cdot 7}{13 \cdot 12}$$

$M$  = estrazione di 2 lampadine nuove  
il II giorno

$$P(M) = P(M|H_1) P(H_1) + P(M|H_2) P(H_2) + P(M|H_3) P(H_3)$$

$$P(M|H_1) = \frac{7 \cdot 6}{13 \cdot 12}$$

$$P(M|H_2) = \frac{6 \cdot 5}{13 \cdot 12}$$

$$P(M|H_3) = \frac{5 \cdot 4}{13 \cdot 12}$$

$$P(H_1 | M) = \frac{P(M | H_1) P(H_1)}{P(M)} \text{ teo di Bayes}$$

$$X \sim U(0, 2)$$

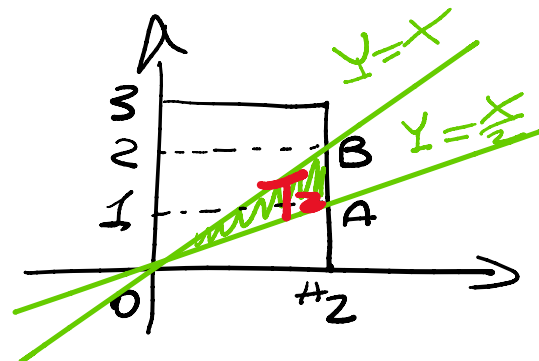
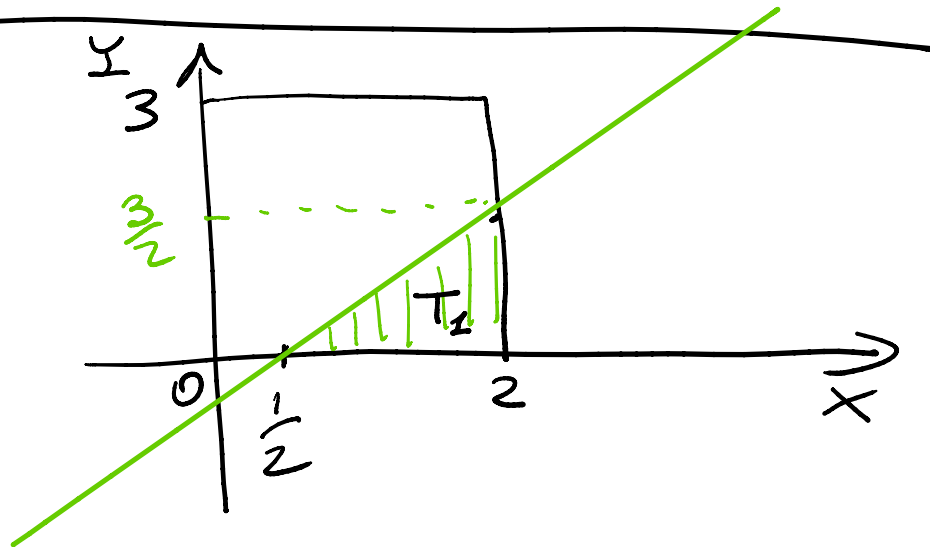
$$Y \sim U(0, 3)$$

$$1) P(X > Y + \frac{1}{2}) =$$

$$= P(Y < X - \frac{1}{2}) =$$

$$= \frac{\text{Area}(T_1)}{\text{Area}(\text{Rektangulo})} =$$

$$= \frac{\left(\frac{3}{2}\right)^2 \cdot \frac{1}{2}}{2 \cdot 3} = \frac{9}{8} \cdot \frac{1}{6} = \frac{3}{16}$$



$$2) P\left(\frac{X}{2} < Y < X\right) =$$

$$= \frac{\text{Area}(T_2)}{\text{Area}(\text{Rektangulo})} = \frac{\text{Area}(\triangle O B H) - \text{Area}(\triangle O A H)}{3 \cdot 2} =$$

$\begin{matrix} \nearrow 2 \\ \searrow 1 \end{matrix}$

$$= \frac{\frac{2^2}{2} - \frac{2 \cdot 1}{2}}{6} = \frac{1}{6}$$

$$3) Z = X + Y \quad X \sim U(0, 2), Y \sim U(0, 3)$$

$$E[Z] = E[X] + E[Y] = 1 + \frac{3}{2} = \frac{5}{2}$$

$$\underset{\substack{\uparrow \\ \text{i.i.d.}}}{\text{Var}(Z)} = \text{Var}(X) + \text{Var}(Y) = \frac{2^2}{12} + \frac{3^2}{12} = \frac{4}{12} + \frac{9}{12} = \frac{13}{12}$$

$$Z_k \text{ i.i.d. } E[Z_k] = \frac{5}{2}, \text{Var}(Z_k) = \frac{13}{12}$$

$$S_{36} = \sum_{k=1}^{36} Z_k \sim N\left(36 \cdot \frac{5}{2}, \sqrt{36 \cdot \frac{13}{12}}\right) = N(90, \sqrt{39})$$

$$\begin{aligned} P(S_{36} > 95) &= P\left(\frac{S_{36} - 90}{\sqrt{39}} > \frac{95 - 90}{\sqrt{39}}\right) = \\ &= P\left(Z > \frac{5}{\sqrt{39}}\right) = 1 - F_Z\left(\frac{5}{\sqrt{39}}\right) \end{aligned}$$

$$= P(Z > \frac{5}{\sqrt{39}}) = 1 - F_Z\left(\frac{5}{\sqrt{39}}\right)$$

06 dicembre 2014

$H_1$  : scelta dell'I scelta  $P(H_1) = P(H_2) = \frac{1}{2}$

$H_2$  : " "  $\Pi$  "

$F$  = 'funzionamento del sistema  
in serie dopo 8 (1000 h)

$$P(F|H_1) = P(X \geq 8) = e^{-4 \times 8} = e^{-32}$$

$$X \sim E(40 \cdot \frac{1}{10})$$

$$P(F|H_2) = P(Y \geq 8) = e^{-8 \times 8} = e^{-64}$$

$$Y \sim E(40 \cdot \frac{1}{5})$$

$$P(F) = P(F|H_1) \cdot P(H_1) + P(F|H_2) \cdot P(H_2) =$$

$$\begin{aligned}
 P(F) &= P(F|H_1) \cdot P(H_1) + P(F|H_2) \cdot P(H_2) = \\
 &= \frac{1}{2} e^{-32} + \frac{1}{2} e^{-64} = \frac{1}{2} e^{-32} (1 + e^{-32})
 \end{aligned}$$

$$P(H_2|F) = \frac{P(F|H_2) P(H_2)}{P(F)} = \frac{\cancel{\frac{1}{2}} e^{-64} \cdot \cancel{\frac{1}{2}}}{\cancel{\frac{1}{2}} e^{-32} (1 + e^{-32})} = \frac{e^{-32}}{1 + e^{-32}}$$